Hertz sphere-wall impact

LAMMPS dem03 | FunDEM native CPU / CUDA | 28 October 2026

PASS	All four runs pass the official 1% peak-energy check.

Core comparison

Run	Steps / frames	Peak force (N)	Max. overlap (um)	Measured e	Energy error (%)
CPU dt	3038 / 3039	384.052493	22.680648	1.0000000051	0.00003318
GPU dt	3038 / 3039	384.052493	22.680648	1.0000000051	0.00003318
CPU dt/2	6076 / 6077	384.052425	22.680645	1.0000000003	0.00000397
GPU dt/2	6076 / 6077	384.052425	22.680645	1.0000000003	0.00000397

Analytic reference: peak force 384.052416289 N; maximum overlap 22.680644976 um; e = 1; contact duration 17.116763980 us.

The official check is 0.5 m v_in^2 ~ 0.4 F_peak delta_peak at cumulative step 2138, with 1% tolerance. CPU and GPU match to floating-point precision. At the official time step, maximum position error at three golden checkpoints is 5.21e-18 m; normalized velocity error is 1.68e-13. Halving dt improves peak and energy errors.

Inputs and exact normal-law mapping

Quantity	Input / FunDEM representation
Sphere	Diameter 0.005 m; density 7000 kg/m^3
Initial state / wall	Center (0, 0, 0.002550) m; velocity (0, 0, -3.9) m/s; wall z = 0
Hertz coefficient	LAMMPS k_n = 7.11111e10 Pa; E* = 3 k_n / 4 = 5.3333325e10 Pa
Elastic representation	Both materials: Poisson ratio 0.25; E = 9.9999984375e10 Pa
Damping / other forces	e = 1; automatic damping = 0; explicit override off; all friction, gravity and global damping = 0
Time / output	dt = 1e-8 s; duration 30.38 us; half-dt = 5e-9 s; every step retained

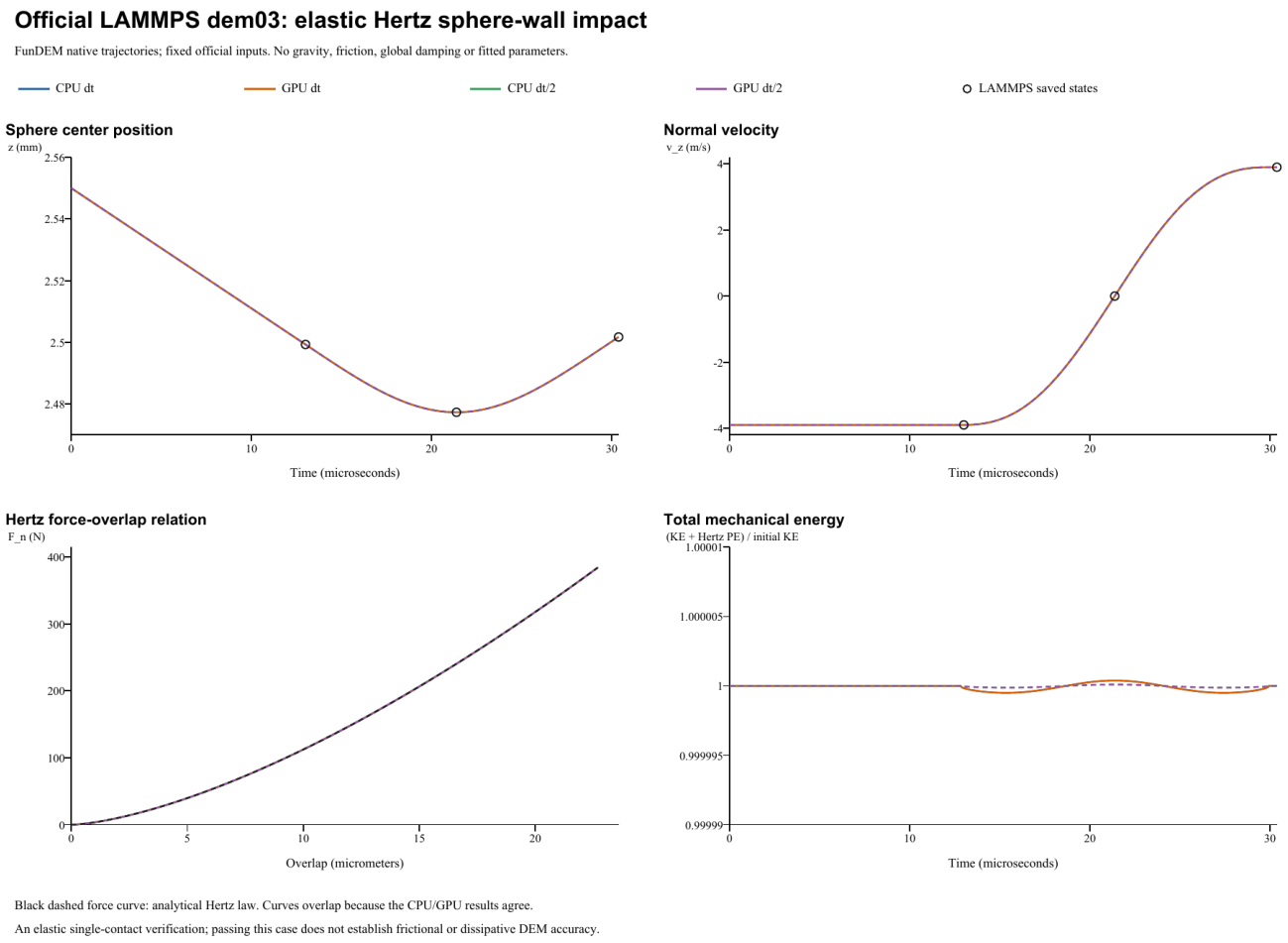
F_n = k_n sqrt(R) delta^(3/2) = (4/3) E* sqrt(R) delta^(3/2). FunDEM combines both elastic compliances, including the infinite-mass wall. Poisson ratio is an equivalent representation choice; E is derived to preserve E*. No fitted parameters. The three consecutive LAMMPS segments are one unchanged FunDEM event.

Reference provenance

Official LAMMPS YAML and golden states; commit f9720ce2d176d615495d1d6c31349b1f71be1c73. Local YAML Git blob: 8ab4e654a5c4612846413c65d239c37cf695ef4d. LAMMPS was not rerun locally. The reference uses the pinned official states plus an independent Hertz solution. The YAML epsilon = 1e-10 is a LAMMPS regression setting, not a universal cross-code tolerance.

Curves and retained evidence

Official checkpoints, Hertz analytic relation and real native trajectories



Open markers are official saved states. The dashed curve is the analytic Hertz force-overlap relation. Overlapping native CPU/GPU curves are expected. Analytic values were not used to drive or correct the simulated motion.

Complete native results are retained

CPU dt: run-20261028-031141-411-f6f00539-bf7f-4746-a07d-8dcd059ff0e5

GPU dt: run-20261028-031207-487-7e0a855b-7969-440f-8dbe-13786b5b1039

CPU dt/2: run-20261028-031321-869-a8ed2326-28ac-45ac-854f-29b169df9c6f

GPU dt/2: run-20261028-031411-746-3a003f4a-3a58-4283-9df4-8784b6ff55c1

Run manifest | Parameter mapping | Analysis JSON | Native archive ZIP. Result folders retain run.json, frames and contacts; trajectory.csv and native-audit.json are kept for each project.

Execution and validation limits

Real SimulationSession runs used linked FunDEM 0.4.3 CPU/CUDA native libraries; no independent collision integrator or contact solver changes were used. Build version and hashes are recorded in tools/native-build-manifest.json and tools/native-build-hashes.json. Byte identity with the installed GUI executable was not established.

Verified scope: frictionless elastic normal Hertz contact. Dissipative restitution, tangential friction, rolling resistance and many-particle flows require separate tests.